

FELIX SHAW - BELL

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PROFILE

- Citizenship: New Zealand
- Current residence: Auckland, New Zealand

A high-achieving and motivated mathematician with aspirations of making a lasting contribution to both mathematics and the wider community. I focus my mathematical abilities on real-world modelling, particularly in problems involving uncertainty. My strong academic record is complemented by industry experience, allowing me to apply myself in both a rigorous and practical manner.

EXPERIENCE

Graduate Teaching Assistant (part-time) <i>University of Auckland</i>	March 2026 - June 2026 <i>Auckland, New Zealand</i>
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- Delivered tutorials to classes of 20 to 40 students for first-year mathematics courses.

Data Scientist (full-time) <i>Shift Finance</i>	February 2025 - August 2025 <i>Sydney, Australia (Remote)</i>
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- Developed a machine learning model to assist and automate credit risk assessments.
- Consolidated multiple data sources into SQL tables, now widely used within the company.

Mathematics Research Scholar (full-time) <i>University of Auckland</i>	November 2024 - February 2025 <i>Auckland, New Zealand</i>
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- Worked under Professor Simon Harris and Dr Jesse Goodman to study branching processes.
- Funded by the University of Auckland Summer Research Scholarship.

Data Analyst Intern (full-time & part-time) <i>Shift Finance</i>	November 2023 - November 2024 <i>Sydney, Australia (On-site & Remote)</i>
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- Built and deployed a company-specific Python library providing a machine learning pipeline.
- Contributed to the development of Shift's credit risk models.
- Carried out feature engineering work to enhance model performance from existing datasets.

EDUCATION

MSc, Applied Mathematics (awaiting results) <i>University of Auckland</i>	July 2025 - July 2026 <i>Auckland, New Zealand</i>
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- Key areas of study: stochastic modelling, tipping point systems, approximation methods.
- Masters dissertation: '*Eigenmode approximations of noisy tipping-point systems*', supervised by Professor Graham Donovan. The project applied eigenmode approximations of the Fokker-Planck equation to estimate the risk of tipping in tipping point systems.

BSc (Hons, 1st Class), Applied Mathematics <i>University of Auckland</i>	February 2021 - November 2024 <i>Auckland, New Zealand</i>
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- Key areas of study: mathematical modelling, numerical methods, machine learning.
- Honours dissertation: '*Stochastic modelling of soil nitrogen in intensive agriculture*', supervised by Dr Steve Taylor. The project developed a system of stochastic differential equations to model the important issue of nitrogen pollution into agricultural waterways.

SKILLS

- Technical: MATLAB, Python, SQL, $\text{\LaTeX}$.
- Languages: English (native), Spanish (elementary).

CONFERENCES AND WORKSHOPS

New Zealand Mathematics and Statistics Postgraduate Conference November 2025
'Eigenmode approximations of noisy tipping-point systems' (presentation)

AWARDS AND HONOURS

- University of Auckland Research Masters Scholarship (2025, stipend and free tuition).
- University of Auckland Summer Research Scholarship (2024, stipend).
- University of Auckland Young Scholars Program (2020).
- NZQA Calculus Scholarship (2020).
- NZQA Physics Scholarship (2020).
- Rotary Science & Technology Forum Sponsorship (2019).